

Indian Statistical Institute

BSDS IInd Year

Academic Year 2025 - 2026: Semester I

Course: Probability II

Instructor: Antar Bandyopadhyay

Assignment # 10

Date Given: October 29, 2025

Date Due: November 07, 2025
Total Points: 10

1. Let $(X_n)_{n \geq 1}$ be a sequence of *non-negative* random variables such that, $X_n \downarrow X$, where X is another random variable. Assume that $X_1 \in \mathcal{L}_1$, then show that, $X_n \in \mathcal{L}_1$ for all $n \geq 1$ and $X \in \mathcal{L}_1$. Further, show that $X_n \xrightarrow{\mathcal{L}_1} X$.
2. Suppose $\{X_n\}_{n \geq 1}$ is a sequence of independent random variables, such that, $X_n \sim \text{Poisson}(\lambda_n)$, where $\lambda_n := \log\left(1 + \frac{1}{n}\right)$, $n \geq 1$. Then show that $X_n \xrightarrow{\mathcal{L}_2} 0$, but $\{X_n\}_{n \geq 1}$ does not converge to 0 almost surely.
3. Suppose $X_1, X_2, \dots, X_n, \dots$ be *i.i.d.* random variables taking only ± 1 values with equal probabilities. Let $S_n := X_1 + X_2 + \dots + X_n$ where $n \geq 1$. Show that

$$\frac{S_n}{n} \xrightarrow{\mathcal{L}_2} 0.$$

[Recall, by **WLLN** $\frac{S_n}{n} \xrightarrow{\mathbf{P}} 0$ and by **LLN** $\frac{S_n}{n} \longrightarrow 0$ a.s.]

4. Let $p > 1$, show that there will always exist a sequence of random variables, say $\{X_n\}_{n \geq 1}$ with each $X_n \in \mathcal{L}_p$, such that, $\|X_n\|_p \longrightarrow \infty$ as $n \rightarrow \infty$. In other words, $\mathcal{L}_p$ is *unbounded*.